CREMA

"The essence of the coffee bean"

DESIGN MILESTONES

Gizem Yıldırım

RESEARCH

Interview Insights

Promt: Can you summarize the key points of these interviews I conducted, the areas where users struggled and where they were happy?

Role of AI: Clarifier
AI Output:
Key Points:

- Small kitchens = real constraint (Interview 1 & 2)
- Even large kitchens avoid clutter (Interview 3)
- Countertop visual crowding is as important as physical space
- Storage capacity matters as much as counter size
- Small machines are preferred because they "fit everywhere"

Core Intent:

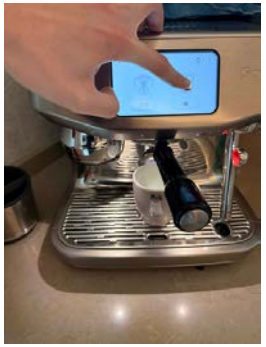
- Compact footprint
- Default simplicity
- Optional depth
- High-quality coffee experience

Problems:

- Large counter space usage
- Visual clutter
- High cognitive load
- Expensive initial investment
- Overwhelming for beginners

Observation Findings

- My observations revealed that the coffee quality was better with semi-automatic machines. The issue of them taking up too much space on the counter was more apparent in my second observation.
- Semi-automatic espresso machines take longer to grind espresso compared to fully automatic capsule espresso machines.
- Additionally, the Nespresso user said that its speed and compact size were advantages.



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DEFINE : PERSONA

Promt: Can you analyze these interviews and find the bullet points of them? Find their similarities and differences and based on these can you create a persona please.

Role of AI: Clarifier
AI Output: Emre is a design-conscious urban professional who is transitioning from convenience coffee systems to specialty brewing. He seeks a modular, adaptable machine that evolves with his skills and space without overwhelming him with professional complexity.

PERSONA BOARD

Name: Emre Kaya
Age: 29
Occupation: Architect (Hybrid Worker)
Location: Urban Apartment
Lifestyle: Design-conscious, evolving coffee enthusiast

About Emre • Lives in a rented apartment with limited counter space • Invests in long-term, high-quality products • Enjoys slow morning routines before work • Values minimal, clean aesthetics • Learns about coffee through online content	**Coffee Journey** • Started with capsule systems • Transitioning into specialty coffee • Buys fresh beans from local roasters • Wants control without technical overload • Seeks a system that evolves with his skills
Pain Points • Capsule waste feels unsustainable • Professional machines are intimidating • Small kitchens limit large appliances • Replacing full machines feels wasteful	**Needs & Goals** • Modular system that adapts over time • Simple and intuitive interface • Compact and spatially efficient design • Upgradeable components (grinder, milk module, etc.) • Long product lifespan

IDEATE : EVALUATION OF IDEA

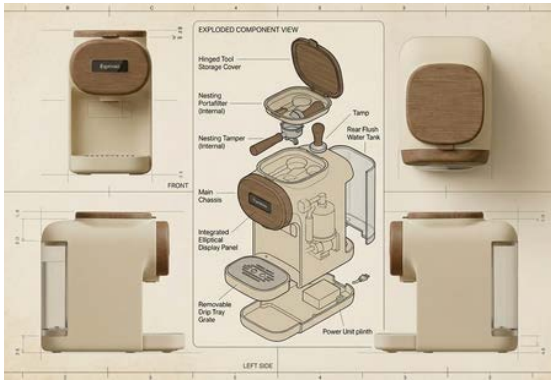
Role of AI: Expander

When I verbally expressed my ideas to the AI, it was able to give me visual results. Of course, where these results weren't satisfactory, I used applications like Photoshop to feed the AI with images that better reflected the product I wanted, thus achieving better results.

Initially, after gathering user reviews and conducting research, I realized there were some specific problems with semi-automatic espresso machines: **they took up too much space on the countertop, were very complex, and fully automatic machines produced poor coffee quality**. My initial idea was to focus on a modular coffee machine, but as I continued with the project, I realized I was straying from the scope and focusing too much on accessories. For this reason, and based on feedback from my instructors, I decided that it would be better to focus solely on the espresso machine itself.

IDEATE : EVALUATION OF IDEA

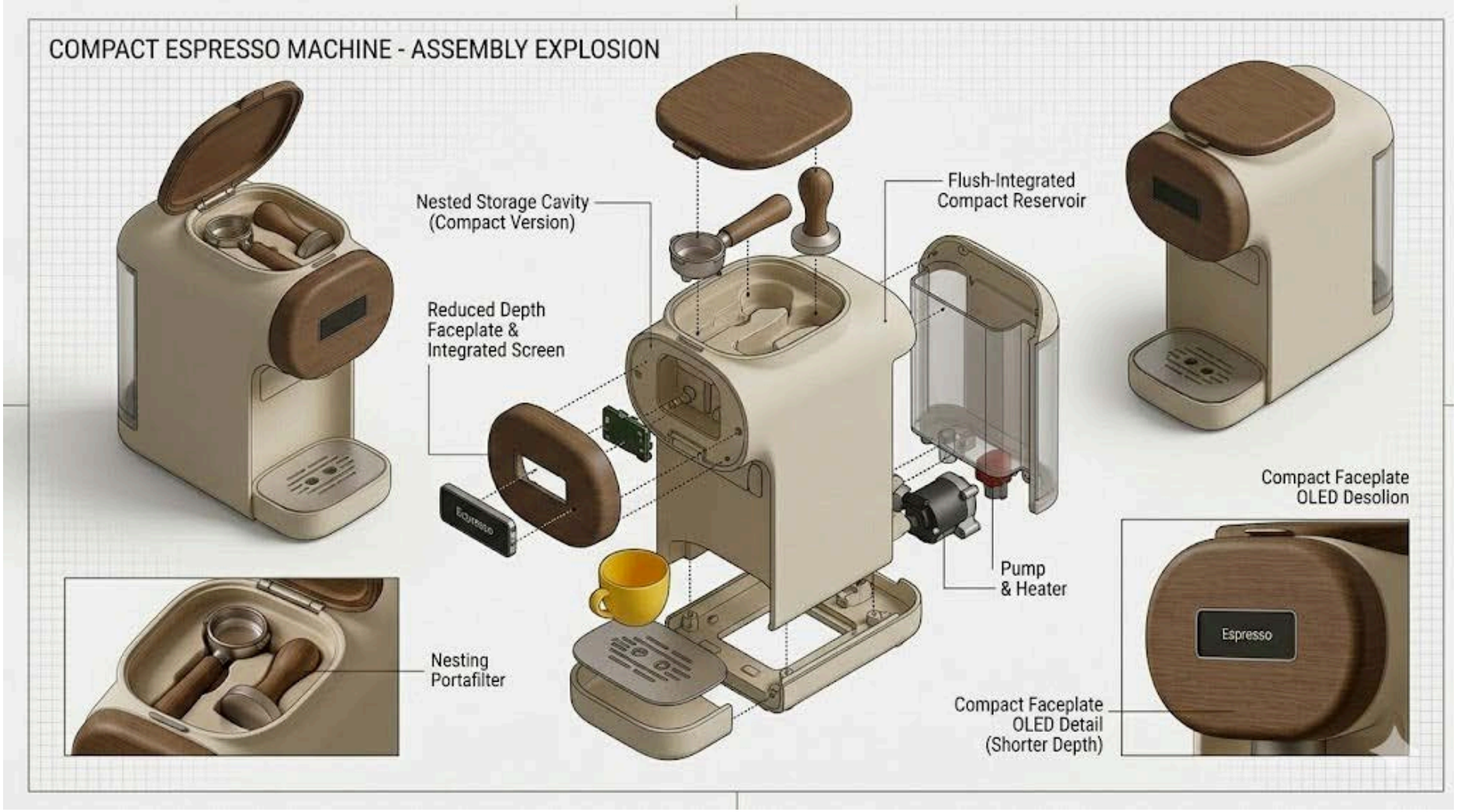
Role of AI: Critique
The AI provided critical feedback on my design, identifying areas for improvement through pain points. It also pointed out oversights, such as potential damage to the wood materials from steam exposure, and helped me find solutions. Furthermore, it guided me in material selection with suggestions on various coating options.



As a result, I designed a **minimalist, nature-inspired semi-automatic coffee machine that takes up minimal countertop space and is easy to use**. In this way, I found a solution to all the problems I identified in my research within a single product.

PROTOTYPE - TEST : TECHNICAL DETAILS

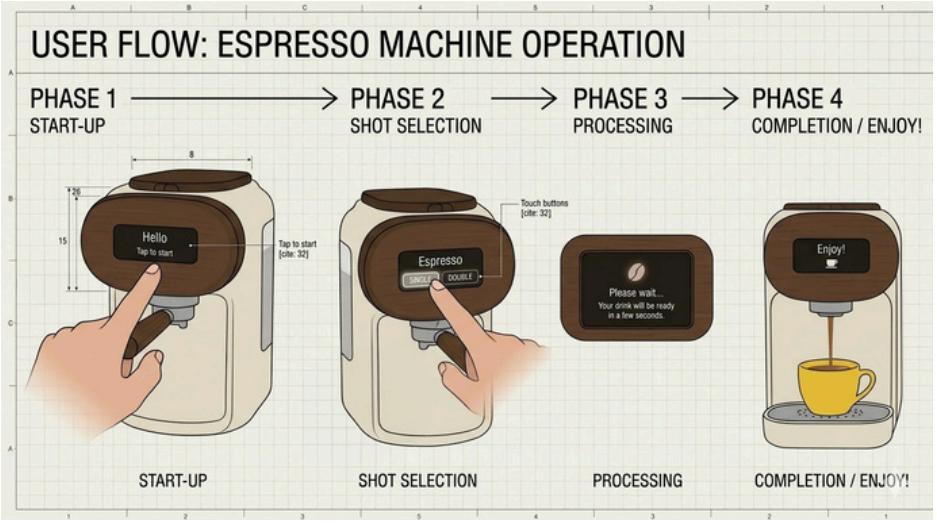
Role of AI: Critique
AI provided me with critical feedback on ergonomic improvements, helping me to incorporate ergonomic enhancements into my design.



PROTOTYPE - TEST : TECHNICAL DETAILS

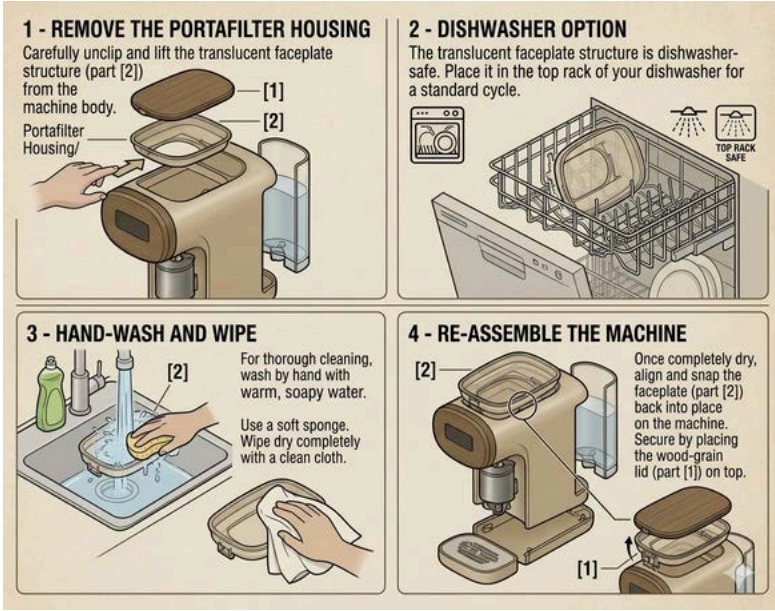
Role of AI: Critique

Display Details



For the display, I designed a user-friendly, uncluttered interface that everyone can easily use. By offering two options on the display, I created a simple interface that doesn't confuse the user, allowing them to brew coffee according to their preference without having to waste time understanding it. In addition, the user is informed about the coffee brewing stages with the help of the display.

Cleaning Details



I designed the portafilter so that it can be easily attached and removed from the bed, allowing the user to clean it if something spills. Additionally, I chose silicone for easy cleaning, making the design suitable for both dishwasher and hand washing.

PROTOTYPE : 3D MODEL

Role of AI: Modeling
I uploaded all angles of my design to the artificial intelligence and had it create a 3D model. I used detailed images and very detailed prompts for this.

Before

After

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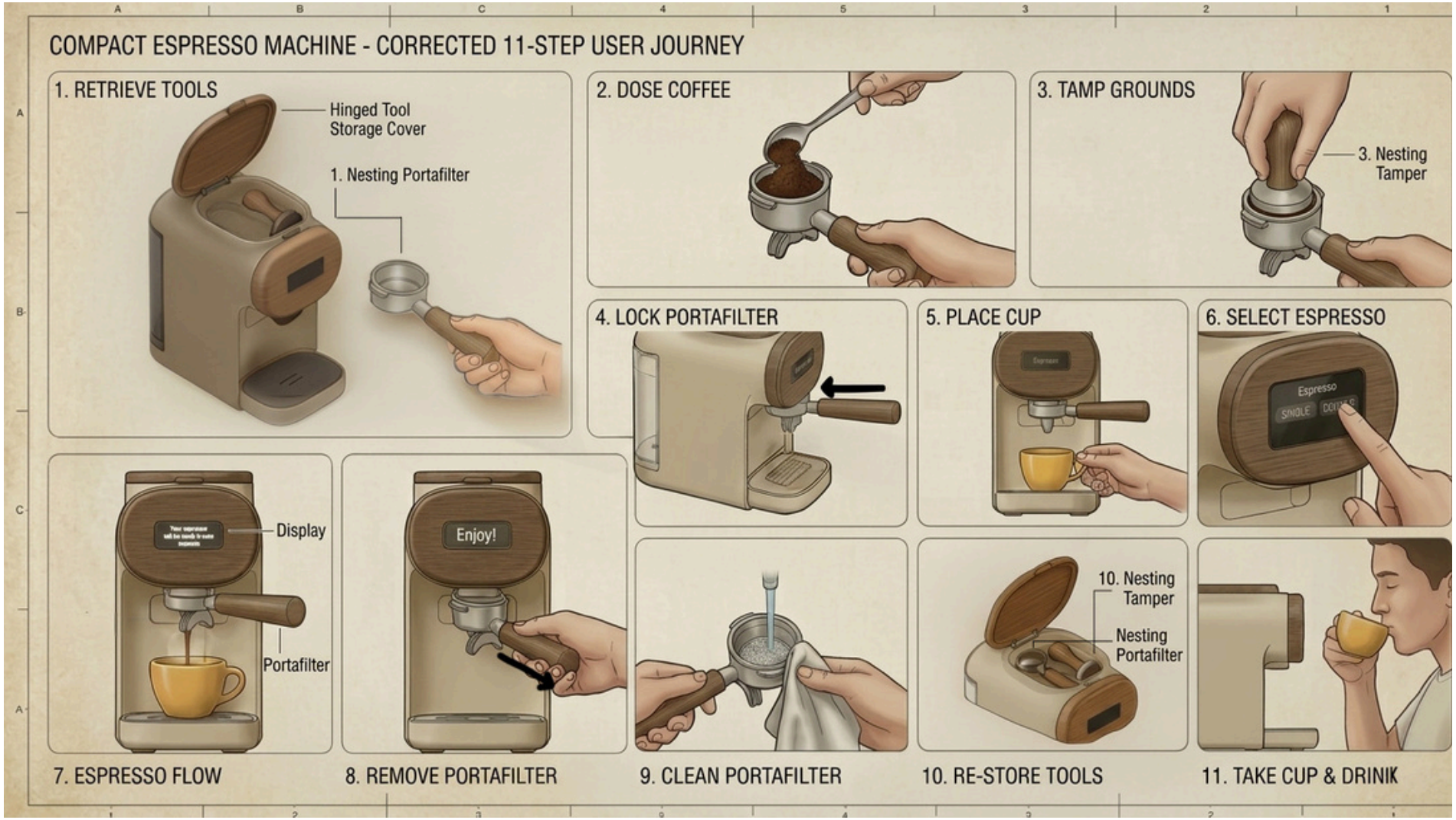
PROTOTYPE : RENDERS

Role of AI: Visual creation
By uploading an image of the product I designed to AI tools and entering the details of the kitchen I wanted, it helped me create photo-quality renders in different locations.

COMMUNICATION : USAGE SCENARIO

Role of AI: Visual creation

After writing the scenario I wanted to the AI, it generated a user scenario for me, but the usage scenario it created messed up my design, and when I asked it to fix it, it was unsuccessful. I made some adjustments in Photoshop to get it to its final form.



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